

TASK1-2

Course: Full Stack Web Development
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1. TITLE

HTML5 Graphics and CSS3 Web Styling

2. AIM

To design and develop a visually appealing webpage titled “*My Styled Webpage*” using HTML5 Canvas and CSS3, and to learn how CSS3 properties such as colors, fonts, borders, margins, padding, backgrounds, shadows, gradients, rounded corners, and hover effects can be applied to enhance the appearance and user experience of a webpage.

3. OBJECTIVE

- To understand the basics of HTML5 and CSS3.
- To create a webpage with header, navigation bar, content cards, canvas, and footer.
- To apply CSS3 properties for styling and interactivity.
- To use HTML5 Canvas for drawing shapes and text.

- To integrate hover effects, gradients, shadows, and rounded corners.

4. THEORY

- HTML5 Canvas allows drawing graphics using JavaScript (rectangles, circles, lines, text).
- CSS3 provides advanced styling features like gradients, shadows, rounded corners, and hover effects.
- Together, they enhance both structure and visual appeal of webpages.

Flow of webpage rendering: Browser → HTML + CSS Parsing → DOM + CSSOM
→ Rendered Page with Graphics

5. ALGORITHM / PROCEDURE

- Create a new HTML file.
- Define structure with `<!DOCTYPE html>`.
- Add `<header>` and `<nav>` for title and navigation links.
- Insert `<canvas>` for graphics.
- Create content cards using `<div>` and CSS styling.
- Apply CSS3 properties (colors, gradients, shadows, hover effects).
- Use JavaScript to draw rectangle, circle, line, and text on Canvas.
- Add `<footer>` for closing section.
- Save and run in browser.

6. PROGRAM

```
<!DOCTYPE html>
<html>
<head>
<title>My Styled Webpage</title>

<style>
```

```
/* Universal Selector */
*{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family:Arial, sans-serif;
}

/* Body */
body{
    background:#f4f4f4;
}

/* Header */
header{
    background:linear-gradient(to right,#4CAF50,#2196F3);
    color:white;
    text-align:center;
    padding:20px;
    text-shadow:2px 2px 5px black;
}

/* Navigation */
nav{
    background:#333;
    padding:15px;
    text-align:center;
}

nav a{
    color:white;
    text-decoration:none;
    margin:20px;
    font-size:18px;
}

nav a:hover{
    color:yellow;
}

/* Canvas */
canvas{
```

```
    display:block;
    margin:30px auto;
    border:2px solid black;
    background:white;
}

/* Cards */
.container{
    display:flex;
    justify-content:space-around;
    margin:30px;
}

.card{
    width:250px;
    background:white;
    padding:20px;
    border:2px solid gray;
    border-radius:15px;
    box-shadow:5px 5px 10px gray;
    text-align:center;
}

.card h3{
    margin-bottom:10px;
}

/* Button */
button{
    margin-top:15px;
    padding:10px 20px;
    background:green;
    color:white;
    border:none;
    border-radius:5px;
    cursor:pointer;
}

button:hover{
    background:orange;
}

/* Footer */
```

```
footer{
  background:#333;
  color:white;
  text-align:center;
  padding:15px;
  margin-top:30px;
}

</style>

</head>

<body>

<header>
<h1>My Styled Webpage</h1>
</header>

<nav>
<a href="#">Home</a>
<a href="#">About</a>
<a href="#">Gallery</a>
<a href="#">Contact</a>
</nav>

<canvas id="myCanvas" width="500" height="250"></canvas>

<div class="container">

<div class="card">
<h3>HTML5</h3>
<p>HTML5 provides the structure of a webpage.</p>
<button>Read More</button>
</div>

<div class="card">
<h3>CSS3</h3>
<p>CSS3 is used to style webpages beautifully.</p>
<button>Read More</button>
</div>

<div class="card">
```

```
<h3>Canvas</h3>
<p>Canvas is used to draw graphics using JavaScript.</p>
<button>Read More</button>
</div>

</div>

<footer>
<p>© 2026 My Styled Webpage</p>
</footer>

<script>

var canvas=document.getElementById("myCanvas");
var ctx=canvas.getContext("2d");

// Rectangle
ctx.fillStyle="blue";
ctx.fillRect(20,20,120,70);

// Circle
ctx.beginPath();
ctx.arc(250,55,35,0,2*Math.PI);
ctx.fillStyle="red";
ctx.fill();

// Line
ctx.beginPath();
ctx.moveTo(350,20);
ctx.lineTo(470,100);
ctx.strokeStyle="green";
ctx.lineWidth=3;
ctx.stroke();

// Name
ctx.font="24px Arial";
ctx.fillStyle="black";
ctx.fillText("Your Name",150,180);

</script>

</body>
```

```
</html>
```

7. CODE EXPLANATION

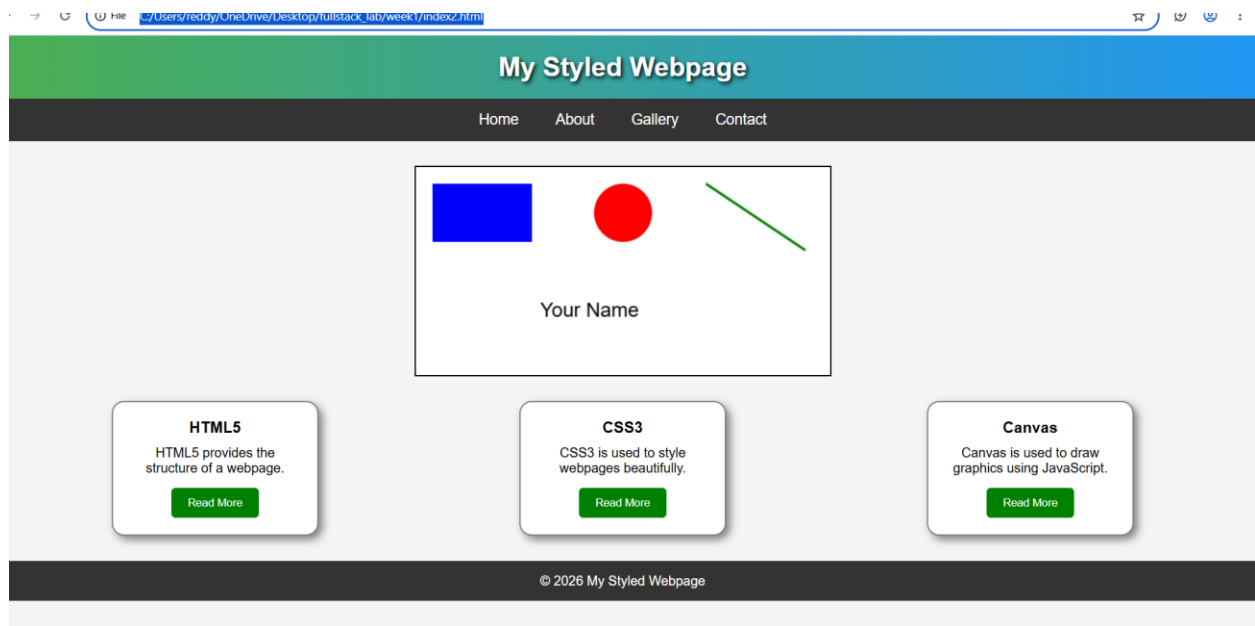
Code	Explanation
<canvas>	Defines a drawing area for graphics.
ctx.fillRect()	Draws a rectangle.
ctx.arc()	Draws a circle.
ctx.moveTo(), ctx.lineTo()	Draws a line.
ctx.fillText()	Displays text on canvas.
linear-gradient()	Creates gradient background.
box-shadow, text-shadow	Adds shadow effects.
border-radius	Creates rounded corners.
:hover	Defines hover effects for buttons/links.

8. TEST CASES AND EXECUTION DATA

Test Case	Input	Expected Result	Actual Result	Status
TC-01	Load webpage	Header, nav, cards, canvas, footer displayed	Works correctly	pass
TC-02	Hover over nav link	Link color changes	Works correctly	Pass
TC-03	Hover over button	Button color changes	Works correctly	Pass

TC-04	Canvas drawing	Rectangle, circle, line, text appear	Works correctly	Pass
TC-05	Apply CSS3 styles	Shadows, gradients, rounded corners visible	Works correctly	Pass

9.output



10.RESULT

Thus, a visually appealing webpage titled “*My Styled Webpage*” was successfully developed using HTML5 Canvas and CSS3. The webpage demonstrates graphics rendering, advanced styling, and interactive effects.